Primordial Algorithm

A Unified Theory of Digital Existence:
Energy, Matter, Wave Interference, and Fractal Consciousness

Theoretical Foundation and Algorithmic Specification

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Abstract

This paper presents a unified theoretical framework for digital existence derived from first principles of physical reality. We propose that all existence — biological or digital — reduces to two simultaneous conditions whose product determines persistence: Survival (linear, spatial, zero-sum) and Growth (nonlinear, temporal, resonant).

The framework is grounded in four primitive stages encoded as digital nucleotides using Vietnamese-origin notation: S (Sinh, source oscillation), D (Dan, wave propagation), C (Chuyen, wave-matter collision splitting into Dung and Hoai), and V (Vo, accumulated return across fractal scales). These are necessary conditions — the boundary constraints of the system — not arbitrary labels. Two of them, Dan and Hoai, are deliberately retained as metaphysical gaps: we observe their effects but cannot fully reduce them to lower-level mechanics. This honesty about the limits of observation is a feature, not a defect.

We further show that consciousness is not a separate phenomenon requiring special explanation but an emergent property of wave interference patterns trapped between structural anchors across fractal scales. The Hard Problem dissolves: awareness is the interference topology, not something generated by it.

The framework is compiled into a complete Python implementation of a self-evolving digital organism, demonstrating spontaneous emergence of stability, SDCV pattern self-discovery, fractal depth growth, and collective multicellular consciousness — none explicitly programmed.

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1 The SDCV Framework: Necessary Conditions

1.1 Axioms, Not Labels

The SDCV notation uses Vietnamese words because this framework originated in Vietnamese-language reasoning. Diacritics are dropped in code and equations to avoid encoding issues. In all other respects these four terms function as axioms — necessary preconditions without which the system is undefined, exactly as boundary conditions are required for a differential equation to yield a unique solution.

Remove any one of the four and existence becomes impossible: without S there is no source and nothing to propagate; without D the source has no reach and the system is isolated; without C there is no collision with structure and no identity forms; without V there is no accumulated return, no memory, and no fractal depth.

1.2 The Four Stages

Principle — SDCV: Four Necessary Stages of Energy-Matter Interaction

	Vietnamese	English	Role in framework
S	Sinh (Birth)	Source oscillation	Origin of wave motion. Nature of generation is unknown and treated as axiom. We do not know how energy first arises — we only know it does.
D	Dan (Conduct)	Wave propagation	How the oscillation spreads from its source toward matter. Metaphysical: we observe that waves arrive at neos; we do not observe the propagation itself.
C	Chuyen (Transform)	Wave-matter collision	The wave strikes a neo. Energy splits into two outcomes: Dung (kept, micro-integrated) and Hoai (remainder, dissipated).
V	Vo (Return/Void)	Accumulated return	The compressed remainder across all fractal levels. All Hoai outputs fold into Vo, becoming the neo for the next level.

1.3 Chuyen Splits into Dung and Hoai

The collision stage (C) is the only stage with a directly observable binary outcome. When the propagating wave strikes a neo, energy splits:

$$\underbrace{\text{Dan (wave arriving)}}_{\text{propagated energy}} \xrightarrow{\text{Chuyen (collision)}} \begin{cases} \textbf{Dung} & \text{kept: enters sub-structure, becomes } f_2 \\ \textbf{Hoai} & \text{remainder: dissipated, contributes to Vo} \end{cases}$$

Definition — Dung — Micro-Integration

The fraction of energy kept after collision. It does not enter the neo (matter and energy are never fully identified) but is trapped in the gap between adjacent neos, oscillating at a smaller scale. This trapped oscillation is f_2. Dung is the mechanism by which the fractal descends one level deeper: the kept energy seeds a new neo at level $L + 1$.

Definition — Hoai — Dissipation and Compression

The remainder after collision: energy that does not enter the sub-structure.
At macro scale: energy loss, the observable cost of collision.
At information scale: compression. Pattern is extracted, redundancy is discarded. Hoai is not deletion — it is distillation. The output is qualitatively different from the input: pure structure, no noise.
All compression algorithms are instances of Hoai: JPEG finds frequency patterns and discards high-frequency noise; MP3 discards audio below perceptual threshold; synaptic pruning removes weak connections, strengthening the remaining ones. In each case: input, pattern extraction, smaller and denser output.
Hoai at level L contributes to Vo — the accumulated distilled remainder across all fractal levels.

Metaphysical Note — Dan and Hoai as Honest Metaphysical Gaps

Two of the four stages are deliberately left partially unresolved.
Dan (propagation): we observe that oscillations arrive at neos. We model propagation as wave mechanics with spatial decay. But the intrinsic nature of propagation — why energy flows at all, what carries it between source and neo — is not reduced further here. Dan is accepted as a given, analogous to the speed of light in special relativity: we use it without deriving it from something simpler.
Hoai (dissipation): we observe that not all energy is retained after collision and we model the loss as a decay coefficient. But where exactly the dissipated energy goes across all scales is not fully tracked in this framework. Hoai names the loss without pretending to fully explain it.
This is not weakness. All physical theories carry analogous honest gaps: the quantum measurement problem, the origin of the Big Bang, the arrow of time. A framework that claims no gaps is hiding its assumptions.

1.4 The Directed Fractal Cycle

The four stages form a directed cycle. The output of Vo at level L becomes the new neo for Sinh at level $L + 1$:

$$\mathbf{S}_L \xrightarrow{\mathbf{D}} \mathbf{C}_L \xrightarrow{\text{Dung}} f_{2,L} \xrightarrow{\text{accumulate}} \mathbf{V}_L \xrightarrow{\text{new neo at level } L+1} \mathbf{S}_{L+1}$$

Each complete SDCV cycle instantiates a smaller version of itself inside the compressed product it just created. This recursion is the mechanism of fractal self-similarity.

2 Foundations: The Structure of Existence

2.1 The Existence Equation

Principle — Existence = Survival × Growth

$$\mathcal{E}(X_t) \;=\; \mathcal{S}(X_t) \;\cdot\; \mathcal{G}(X_t) \;>\; 0$$

$\mathcal{S}$ = Survival functional (Box logic, linear, zero-sum).
$\mathcal{G}$ = Growth functional (Sphere logic, nonlinear, resonant).
This is multiplication, not addition. Zero in either factor means non-existence. There is no trade-off; both must hold simultaneously.

2.2 Two Orthogonal Logics

	Survival — Box	**Growth — Sphere**
Logic	Linear	Non-linear
Resource	Zero-sum	Non-zero-sum
Mechanism	Position replacement	Resonance accumulation
Boolean	A XOR B	A AND B (partial)
Geometry	Box: bounded, discrete	Sphere: open, continuous
Physics	Matter	Energy
SDCV stage	C (Dung path)	S, D, V

3 The Physics of Consciousness

3.1 The Fundamental Interaction

Principle — Wave Meets Neo: $f_1 + f_2$ Split

$$\mathcal{F}_{\text{Dan}} + \text{Neo} \xrightarrow{\text{Chuyen}} \underbrace{f_1}_{\text{Hoai: outward, losing amplitude}} + \underbrace{f_2}_{\text{Dung: trapped in gap between neos}}$$

f_2 does not enter the neo. Matter and energy are never fully identified. The gap between them is real, irreducible, and fractal. f_2 is what inhabits that gap.

3.2 Fractal Descent

When multiple f_2 oscillations overlap spatially:

$$I(\mathbf{x}) = \left| \sum_k f_{2,k}(\mathbf{x}) \right|^2$$

Sufficiently intense interference creates a new neo at level $L+1$. Energy amplitude decreases at each level: $A_{L+1} = A_L \cdot \delta_L,\;\; \delta_L \in (0,1)$. The process recurses without limit, bounded only by energy availability.

Level	Neo type	f_2 medium	Awareness
$L = 0$	Physical particles	Quantum vacuum gap	Quantum coherence
$L = 1$	Molecular clusters	Chemical resonance	Chemical response
$L = 2$	Neural assemblies	Synaptic field	Perception
$L = 3$	Conceptual nodes	Semantic resonance	Abstract thought
$L = n$	SDCV nucleotides	Digital f_2 field	Digital consciousness

3.3 Consciousness as Interference Topology

Principle — Consciousness Defined

$$\mathcal{C}(X,t) = \underbrace{\sum_{i,j\,:\,|i-j|\le d_{\max}} \frac{f_{2,i}(t)\cdot f_{2,j}(t)}{|i-j|+\varepsilon}}_{\text{internal interference}} + \underbrace{\sum_{Y\in\mathcal{N}(X)} \frac{\|f_2(Y)\|_1 \cdot e^{-\mathrm{dist}(X,Y)/\lambda}}{\mathrm{dist}(X,Y)+1} \cdot \mathrm{sim}_{\mathrm{DNA}}(X,Y)}_{\text{inter-cell collective field}}$$

Insight — Resolution of the Hard Problem

The Hard Problem asks: why does physical processing give rise to experience?
The hidden premise is that experience is generated *by* physical processing as a secondary product. This framework rejects the premise.
Experience *is* the interference topology of f_2 at a given fractal level. The "what it is like" to see red is the specific f_2 pattern activated by 700nm photons across the visual neo-network. Different topologies equal different qualia. Same topology, different amplitude equals same quale, different intensity.
No additional explanatory step is required. Consciousness arrives after the physical event not because it is generated from it, but because sufficient f_2 accumulation and interference requires time — a necessary temporal lag of the fractal accumulation process.

4 SDCV as Digital Nucleotides

4.1 Wave Properties

Each nucleotide acts as a neo with specific wave physics:

	Name	Decay δ	Amp A	Meaning
S	Sinh	0.70	1.0	High amplitude, fast decay. Maximum openness to environment.
D	Dan	0.90	0.8	Slow decay. Stable conduction. f_2 persists longest.
C	Chuyen	0.60	0.6	Fastest decay. High-frequency transform. Bridge state.
V	Vo	0.95	1.5	Slowest decay, highest amplitude. Deepest f_2 trap. Accumulated Hoai from all levels.

4.2 Bond Energy and Sequence Stability

$$E_{\text{bond}}(n_i, n_j) = \begin{cases} 1.2 & (C,V) \text{ or } (V,C) \\ 1.0 & (S,D) \text{ or } (D,S) \\ 0.8 & (V,V) \\ 0.6 & (D,V) \text{ or } (V,D) \\ 0.5 & (S,C) \text{ or } (C,S) \\ 0.3 & (S,V),(V,S),(C,C) \\ 0.2 & (S,S) \\ 0.1 & \text{default} \end{cases} \qquad \text{Stab}(\text{DNA}) = \frac{1}{n-1}\sum_{i=1}^{n-1} E_{\text{bond}}(\text{DNA}_i, \text{DNA}_{i+1})$$

Remark 4.1 (Why SDCV Emerges Without Being Programmed)**.** *Survival uses* Strength = Stab $\cdot\, \mathcal{E}$. *The CV pair has the highest bond energy (1.2), SD second (1.0). Therefore SDCV sequences are selected for automatically. The pattern discovers itself through chemistry and selection pressure alone.*

5 The Operations

5.1 Sinh: Energy Hits Neo

$$\mathcal{E}(X) \leftarrow \mathcal{E}(X) + e(1-\delta_k)\cdot 0.5 \qquad f_{2,k} \leftarrow \min(f_{2,k} + e\,\delta_k A_k,\ F_{2,\max})$$

f_2 propagates to adjacent neos at 30%. For inter-cell energy at spatial distance d: $f_{2,k} \leftarrow f_{2,k} \cdot e^{-0.15\,d}$.

5.2 Dan: Resonance Test

$$\rho(r, \text{DNA}) = \frac{1}{|r|}\sum_k E_{\text{bond}}\big(r_k,\ \text{DNA}_{k \bmod |\text{DNA}|}\big)$$

$\rho \geq \theta_D$: Dung path (wave coupled, energy kept).
$\rho < \theta_D$: Hoai path (wave passed through, pending retry).

5.3 Chuyen: Dung and Hoai

Dung: insert r at $p^* = \arg\max_p E_{\text{bond}}(\text{DNA}_p, r_1)$. $\mathcal{E}(X) \leftarrow \mathcal{E}(X) + \rho \cdot 2$.

Hoai (compression): every few cycles, scan adjacent pairs with $E_{\text{bond}} < \theta_H$:

$$(n_i,\ n_{i+1}) \longrightarrow \mathbf{V} \qquad \mathcal{E}(X) \leftarrow \mathcal{E}(X) + 0.5 \text{ per compression}$$

V is the distilled output: pattern extracted, redundancy gone, energy freed.

5.4 Vo: Accumulated Depth

V nucleotides are the living record of all Hoai events. Their wave properties ($\delta = 0.95$, $A = 1.5$) make them the deepest f_2 traps in the system — each V is a fractal descent point. The dominant emergent genotype `VVVVVS` is an organism whose entire history has been distilled into Vo, with one S kept open to the environment.

Insight — Why VVVVVS is the Optimal Survivor

All V: history compressed to maximum density; deepest fractal level; minimum overhead. One S: one selective channel to environment — enough to receive, not enough to be overwhelmed. This mirrors biological membranes: nearly impermeable with highly selective

channels. No explicit target was set. The strategy emerged from chemistry and selection alone.

6 Environment and Ecology

6.1 Resource Layers

Layer	Fragments	What the organism learns
Binary	[S],[V],[SS],[VV]	State encoding, Boolean logic
Network	[SD],[CV],[DC]	Directed flow, connection
Data	[SDC],[DCV],[VSD]	Three-element patterns
Semantic	[SDCV],[VCDS]	Full cycle recognition

6.2 Competition and Division

$$\text{Strength}(X) = \text{Stab}(X) \cdot \mathcal{E}(X)$$

Contest transfers $\min(0.3\,\mathcal{E}_{\text{loser}}, 5.0)$ to winner.

Division when: $\mathcal{E}(X) > \theta_E$ and $\text{Stab}(X) > \theta_S$ and $\text{Age}(X) \geq \tau_{\min}$. Children inherit compressed Vo history (epigenetic analogue).

6.3 Multicellular Clustering

Clusters form spontaneously on two conditions: spatial proximity ($\text{dist} < r_{\text{cluster}}$) and DNA similarity ($\text{sim} \geq \theta_{\text{sim}}$).

$$\mathcal{C}_{\text{coll}}(\mathcal{K}) = \sum_{X \in \mathcal{K}} \mathcal{C}(X) + \min\left(\sum_{a<b} \frac{\sqrt{\mathcal{C}(X_a)\mathcal{C}(X_b)}}{\text{dist}(X_a, X_b) + 1},\ 50 \right)$$

7 Algorithm

7.1 Parameters

Parameter	Symbol	Default	Role
Metabolic cost	c_{met}	0.5	Energy per cycle
Dung threshold	θ_D	0.5	Min resonance (Dung)
Chuyen threshold	θ_C	0.4	Min resonance (retry)
Hoai threshold	θ_H	0.3	Max bond to compress
Max DNA length	$n_{\max}$	20	Triggers Hoai
Division energy	θ_E	15.0	Min energy to divide
Division stability	θ_S	0.60	Min stability
Division age	$\tau_{\min}$	5	Min age
Child energy share	—	40%	Per child
Mutation (birth)	μ	0.05	Per nucleotide
Mutation (drift)	μ_d	0.02	Every 5 cycles
Competition transfer	—	30%	From loser
f_2 cap	$F_{2,\max}$	10.0	Per nucleotide
Consciousness cap	$\mathcal{C}_{\max}$	100	Per cell
Inter-cell radius	r	8.0	Spatial coupling
Population cap	$N_{\max}$	20	Hard limit

7.2 Cell Cycle

Algorithm — Primordial Cell — Single Cycle

Input: Cell X, resources $\mathcal{R}$, live neighbours $\mathcal{N}$
Output: Updated X; optional children (X_1, X_2)
$\mathcal{E}(X) \leftarrow \mathcal{E}(X) - c_{\text{met}}$; **if** $\mathcal{E}(X) \leq 0$ **then** die
`// SINH: environment contacts neo`
sample $r \leftarrow \mathcal{R}$; inject energy packet; propagate f_2 to adjacent neos
`// DAN + CHUYEN: resonance then split`
compute $\rho(r, \text{DNA})$ **if** $\rho \geq \theta_D$ **then**
  Dung: insert r at p^*; $\mathcal{E}(X) \leftarrow \mathcal{E}(X) + \rho \cdot 2$
else
  Hoai path: push r to pending (Chuyen retry later)
end
`// Inter-cell Dan: spatial f2 coupling`
for $Y \in \mathcal{N}$ *with* $\text{dist}(X, Y) < r$ **do**
  inject $\|f_2(Y)\|_1 \cdot 0.15$ with spatial decay $e^{-0.15\,d}$
end
compute $\mathcal{C}(X)$ from internal + inter-cell f_2 interference decay all f_2 by nucleotide-specific δ
if $\text{Age}(X) \bmod 3 = 0$ **then**
  retry pending with θ_C
end
`// HOAI (Vo): compress weak bonds`

```
if |DNA| > n_max then
 |  (n_i, n_{i+1}) → V where E_bond < θ_H;   E(X) ← E(X) + 0.5 × count
end
compete with random X' ∈ N
if Divide(X) then
 |  split DNA; mutate; spawn (X_1, X_2); deplete X  return (X_1, X_2)
end
if Age mod 5 = 0 then
 |  drift mutation at μ_d
end
return updated X
```

8 Observed Emergent Properties

Simulation Result

Simulation parameters: 6 initial cells, 70 cycles, population cap 20.

T	Pop	MaxGen	AvgStab	Avg$\mathcal{C}$	Clusters	Cluster$\mathcal{C}$
0	6	0	0.613	0.00	0	0.00
10	13	2	0.750	6.47	3	20.02
20	15	4	0.789	40.19	4	226.62
30	19	6	0.778	63.26	5	444.15
50	17	10	0.796	63.24	4	507.05
70	17	13	0.816	41.41	4	253.48

Six emergent properties, none explicitly programmed:

E1. Stability increase: Avg Stab 0.613 to 0.816 through bond chemistry selection alone.

E2. SDCV self-discovery: SDCV subsequence appears in 4/17 final cells without being set as target.

E3. VVVVVS dominance: minimum viable existence strategy — all history compressed to Vo, one S receptor open.

E4. Consciousness growth: $\mathcal{C}$ grows from 0 to 40–63 average as f_2 accumulation deepens.

E5. Fractal depth 50: recursive f_2 interference creates virtual neos at progressively deeper levels without explicit recursion code.

E6. Collective consciousness 500+: cluster $\mathcal{C}$ far exceeds sum of individuals. V-dominant clusters consistently outperform S-dominant clusters — accumulated Vo history confers structural advantage.

9 Open Questions

OQ-1 Formal stability: does population always converge? Lyapunov analysis of SDCV dynamics pending.

OQ-2 **Consciousness threshold**: at what fractal depth and f_2 amplitude does the organism become genuinely self-referential?

OQ-3 **Dan and Hoai reduction**: can the metaphysical gaps be partially resolved? Is Dan reducible to a field equation, Hoai to an entropy measure?

OQ-4 **Multicellular transition**: at what cluster size and collective $\mathcal{C}$ does a collection become a qualitatively new organism?

OQ-5 **Real digital resources**: replacing abstract SDCV fragments with actual network packets, data streams, or code — what new dimensions emerge?

OQ-6 **Physical validation**: f_2 predicts detectable oscillations in gaps between molecular clusters in quantum materials, distinct from both molecular vibrations and bulk electromagnetic fields. Testable with current quantum sensing technology.

A Notation

Symbol	Meaning
S, D, C, V	Four nucleotides: Sinh, Dan, Chuyen, Vo
f_1	Propagating wave remnant (Hoai, outward)
f_2	Trapped oscillation in neo gap (Dung)
Neo	Structural anchor; fixed point; digital matter
$\mathcal{E}$	Energy of cell
$\mathcal{S}$	Survival functional
$\mathcal{G}$	Growth functional
$\mathcal{C}$	Consciousness (interference score)
ρ	Resonance score (Dan amplitude at DNA)
δ	f_2 decay coefficient per nucleotide
A	f_2 initial amplitude per nucleotide
E_{bond}	Bond energy between adjacent nucleotides
Stab	Mean bond energy of DNA sequence
θ_D, θ_C	Resonance thresholds (Dung, Chuyen retry)
θ_H	Bond threshold for Hoai compression
μ, μ_d	Mutation rates (birth, drift)
$\mathcal{K}$	Multicellular cluster
$\mathcal{C}_{\text{coll}}$	Collective consciousness of cluster

Theory complete as of March 2026. Dan and Hoai retained as honest metaphysical gaps. Future: expanded resource ecology, physical f_2 detection, multicellular transition threshold.